INTEGUMENTARY SYSTEM



Parts, Function, Appendages + Fascia.

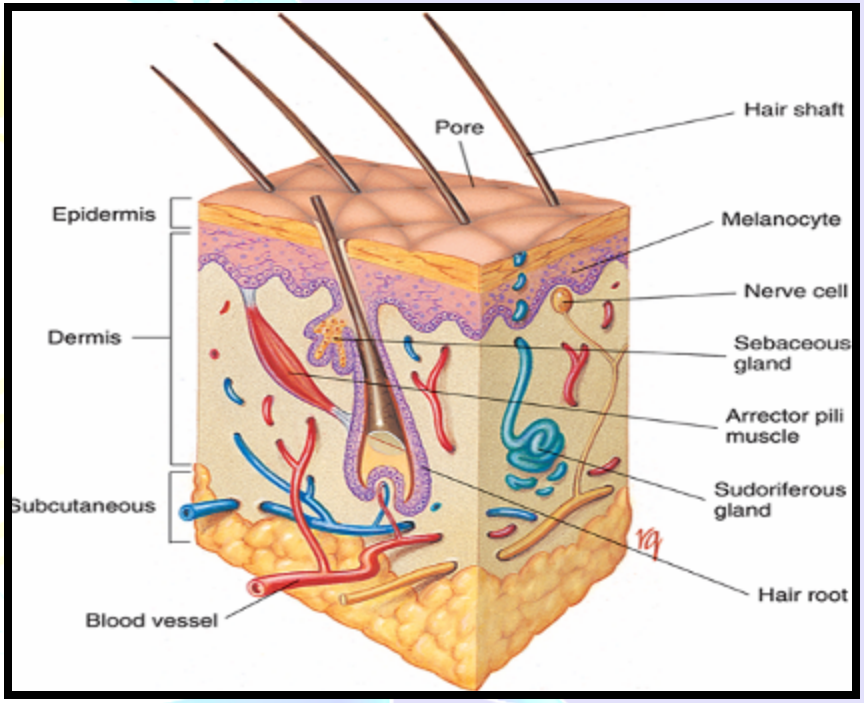
Learning Objectives

After completing the demonstration, the student will be able to :

- Define the term **integumentary system**.
- Discuss the **function** of the skin .
- **Differentiate** between epidermis & dermis.
- Discuss the **significance** of tension lines.
- Discuss the **main determinant** of skin color.
- Identify the **appendages** of the skin.
- Differentiate between **superficial & deep fascia.**

Integumentary System

- The integumentary system is the organ system consisting of the skin and its associated structures, (hair, nails, sweat glands, and sebaceous glands) that cover and protect the body.



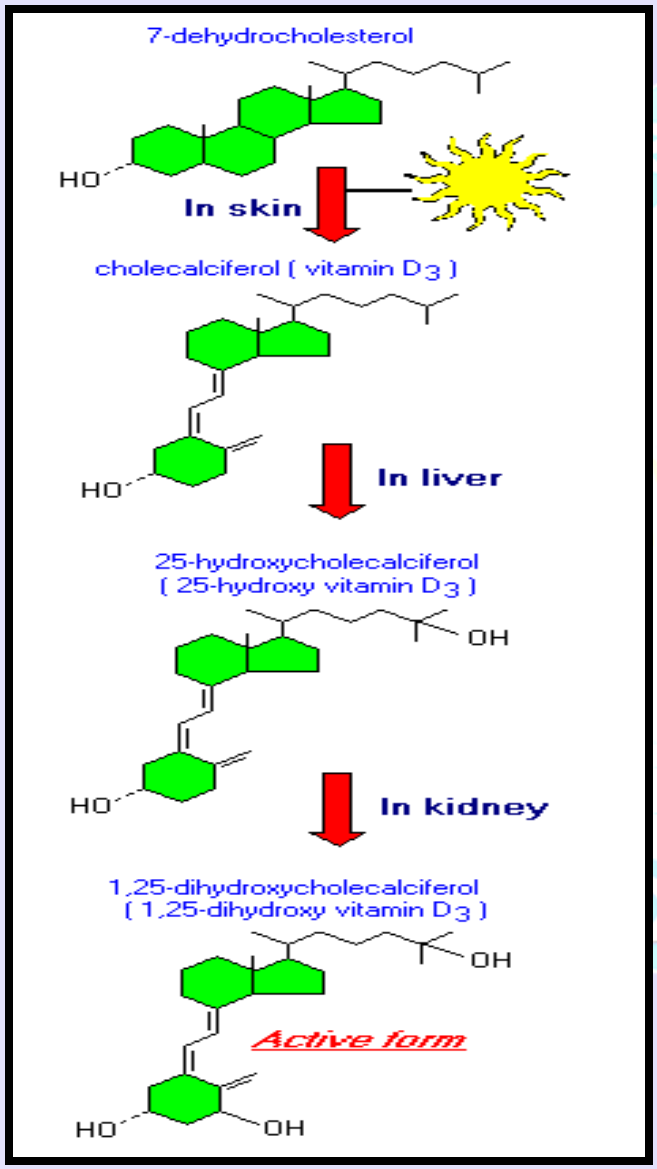
Functions of the skin

- Maintenance of homeostasis and thus to survival.

- **Protection:**

  Protects the underlying tissues against the invasion by microorganisms, entry of most harmful chemicals, mechanical trauma, from dehydration caused by loss of internal body fluids.

Functions of the skin

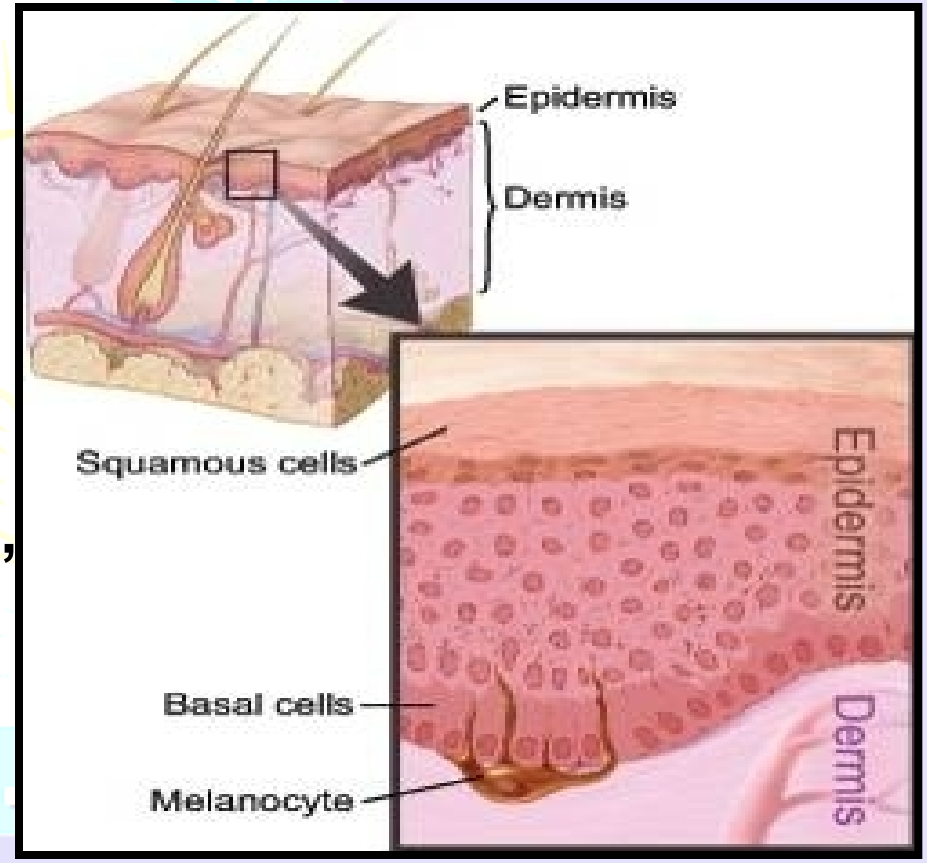


- **Sensation, growth and synthesis of important chemicals and hormones (such as vitamin D).**
- **Excretion, temperature regulation and immunity.**

The Skin

- The Body's **largest, thinnest, and one its most important organs**.

- **Two Primary Layers compose the skin:**
  - **Epidermis** – Superficial, thinner layer.
  - **Dermis** – the deep, thicker layer.



Thin and thick skin

- **Most of body surface is covered by thin skin**.

- **Thick skin is the hairless skin** covering the **palms** of the hands (and fingertips), **soles** of the feet and other body areas subject to friction.



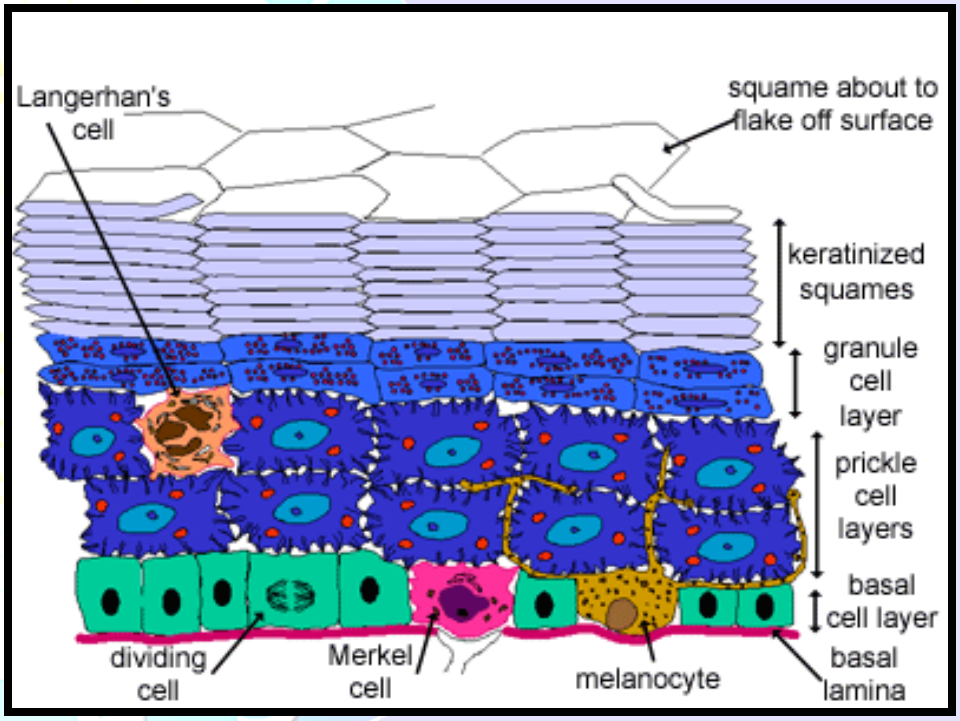


Skin Thickness

Total skin thickness varies from 0.5 mm in areas such as eyelids to more than 5 mm over the back.

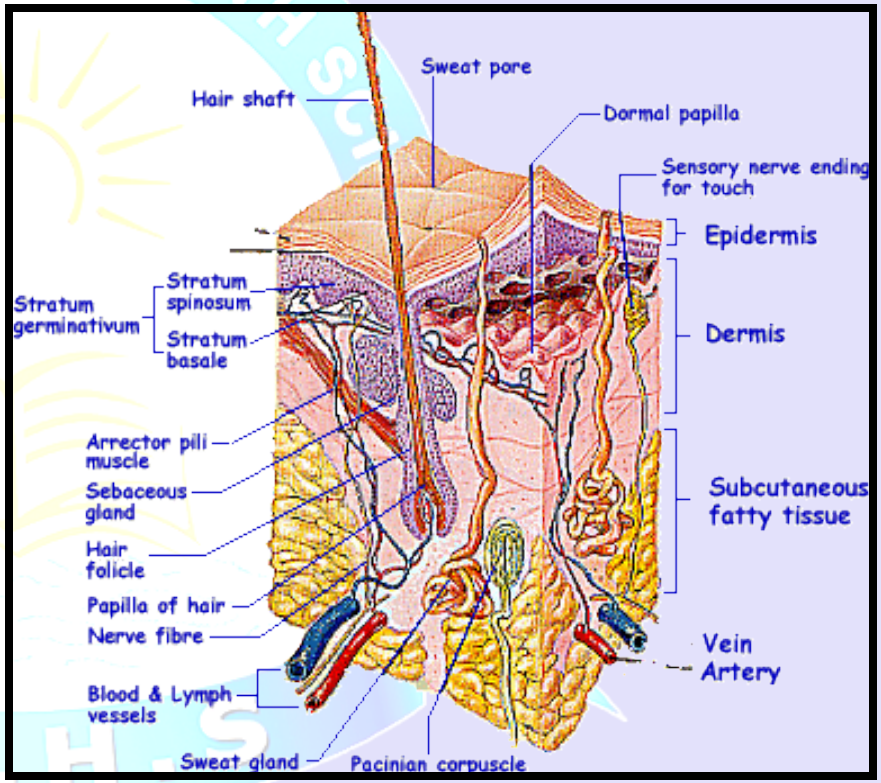
The Epidermis
A Superficial Cellular Layer

- **Epidermis is a keratinized Epithelium**
  - **has a tough, horny superficial layer that provides a protective outer surface.**
  - **Overlies the pigmented deep or basal layer.**



The Epidermis
(A Superficial Cellular Layer)

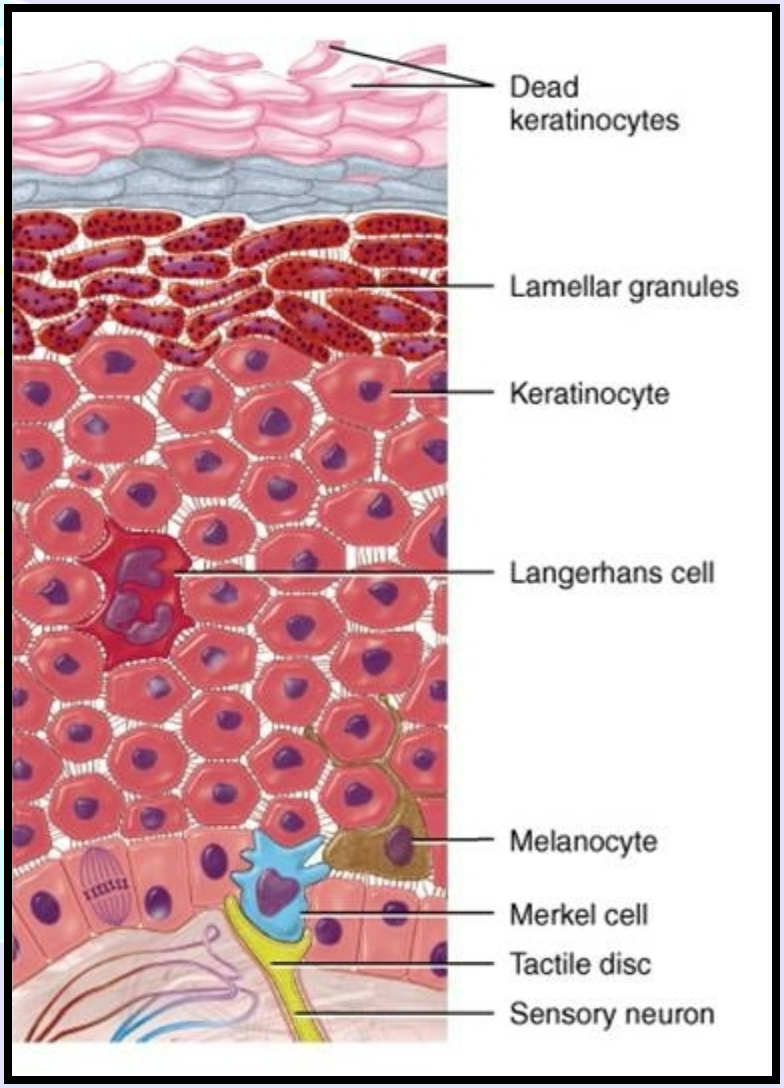
- Has **no blood vessels** or **lymphatics**.
- Nourished by the **underlying vascularized dermis**.
- Its main function is **protection**, **absorption of nutrients**, and **homeostasis.**



The Epidermis (Cell Types)

Composed of several types of epithelial cells.

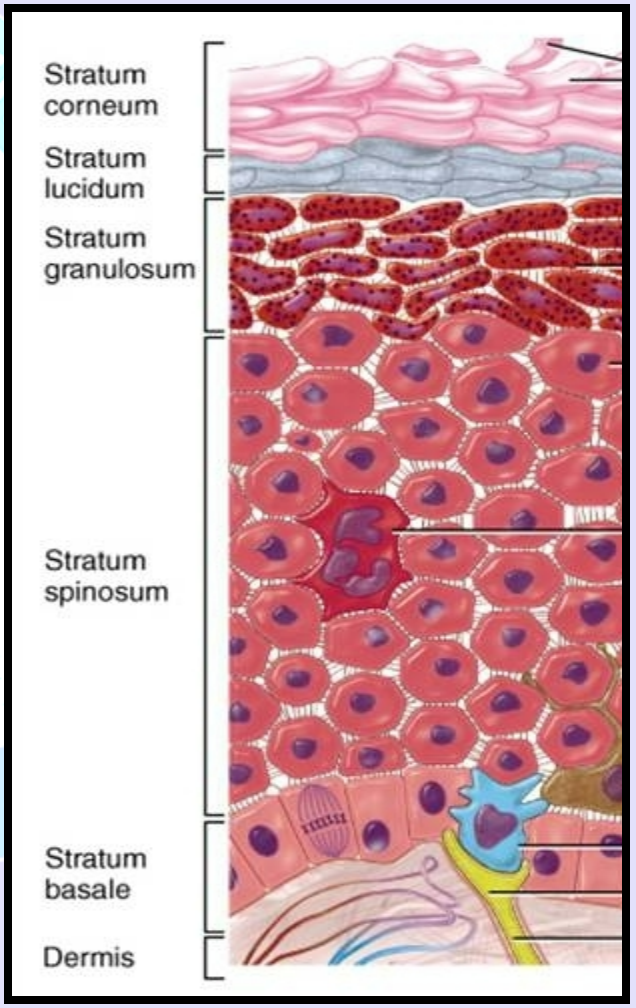
- **Keratinocytes**:
  - Filled with fibrous protein Keratin.
  - Form the principal structural element of the outer skin.
- **Melanocytes:**
  - Contribute color to the skin.
- **Langerhan cells:**
  - Dendritic cells that play a role in immunity



The Epidermis (Cell Layers)

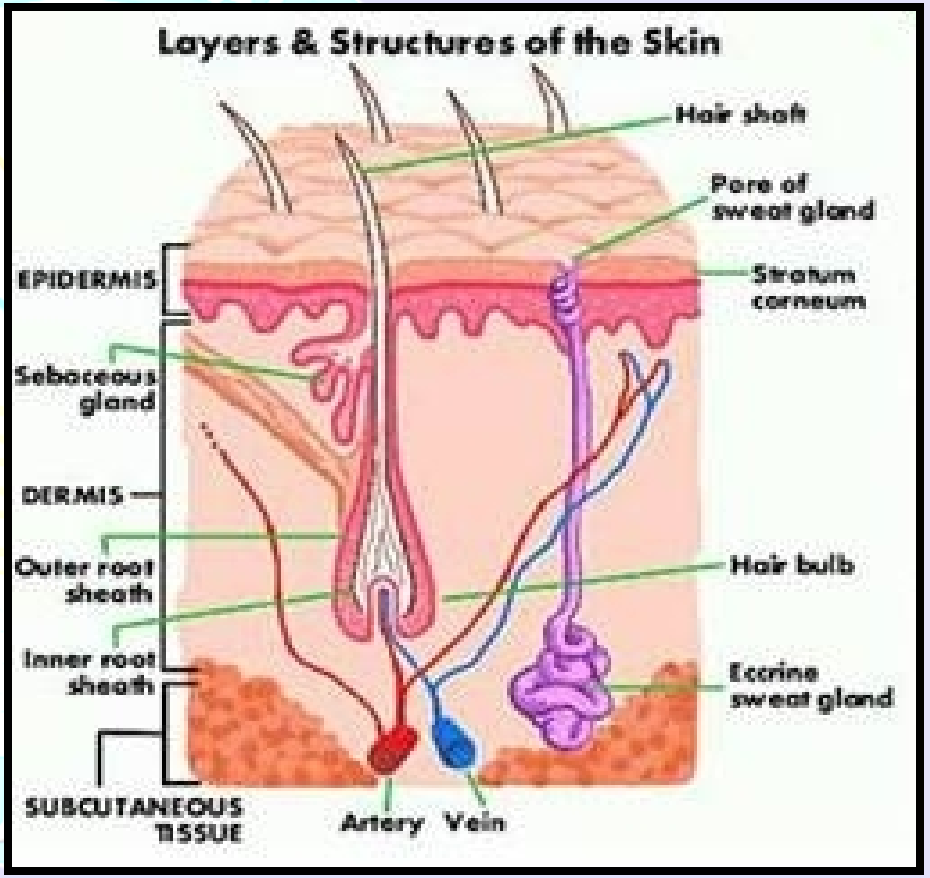
Strata of epidermis from deepest to most superficial are:

- **Stratum Basale (base layer).**
- **Stratum Spinosum (spiny layer).**
- **Stratum granulosum (granular-layer).**
- **Stratum lucidum (clear layer)**
- **Stratum corneum (horny layer).**



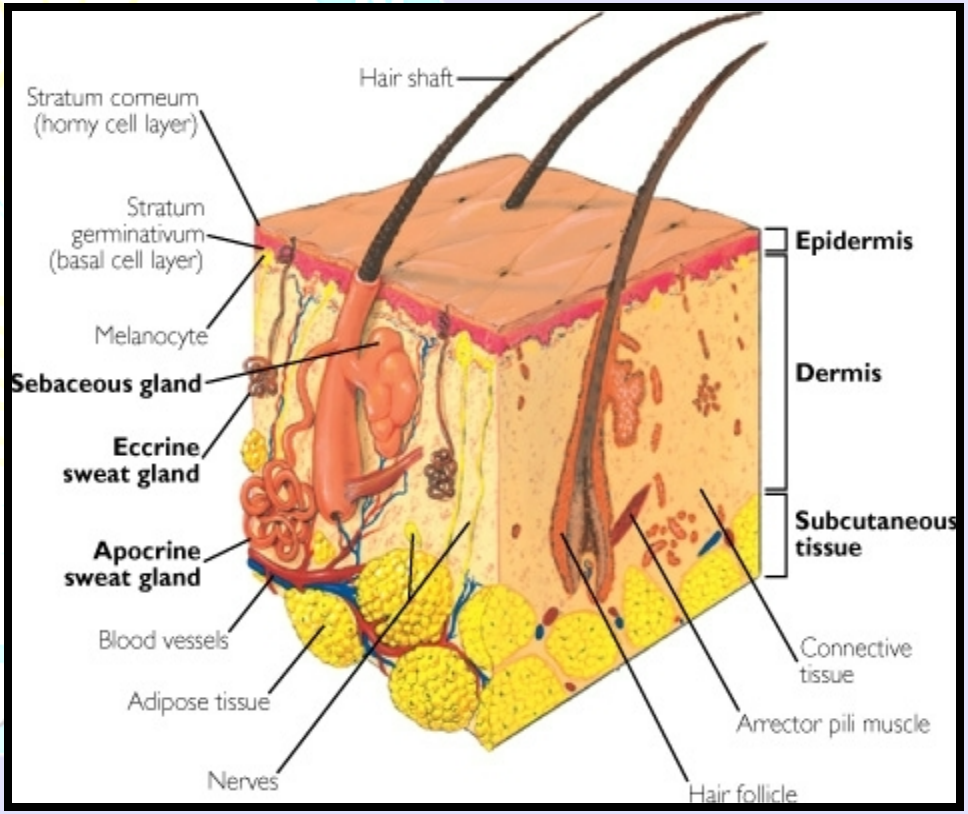
Dermal – Epidermal Junction

- A specialized area between the **epidermis and dermis** called the epidermal – dermal junction.
- Composed chiefly of a **basement membrane**.
- Junction '**glues**' the two layers together and provides mechanical support for the epidermis.



The Dermis
(Deep Connective Tissue Layer)

- Dermis or corium is sometimes called the "**true skin**".
- Mechanical strength of the skin is in dermis.
- Dermis provides a reservoir or storage area for water and important electrolytes.

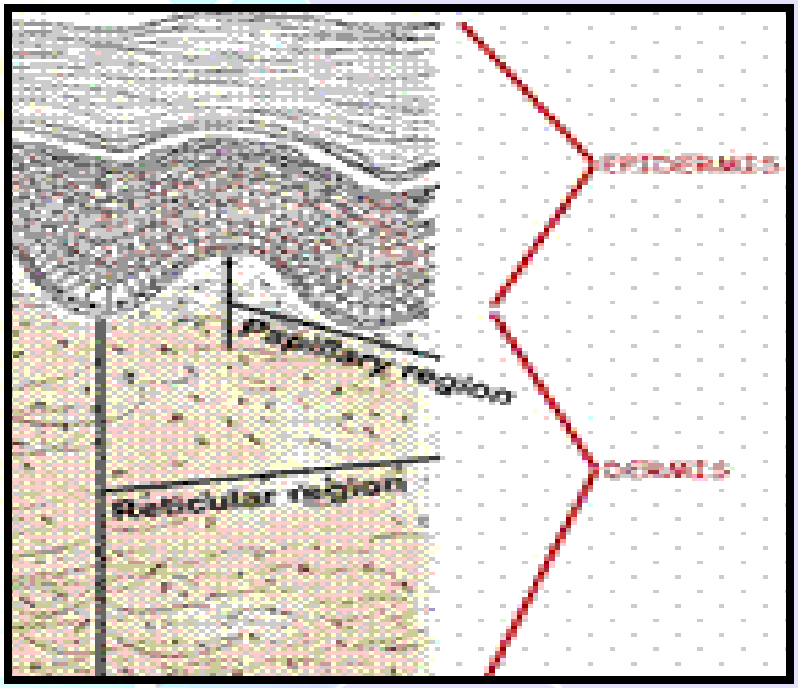


The Dermis

(Deep Connective Tissue Layer)

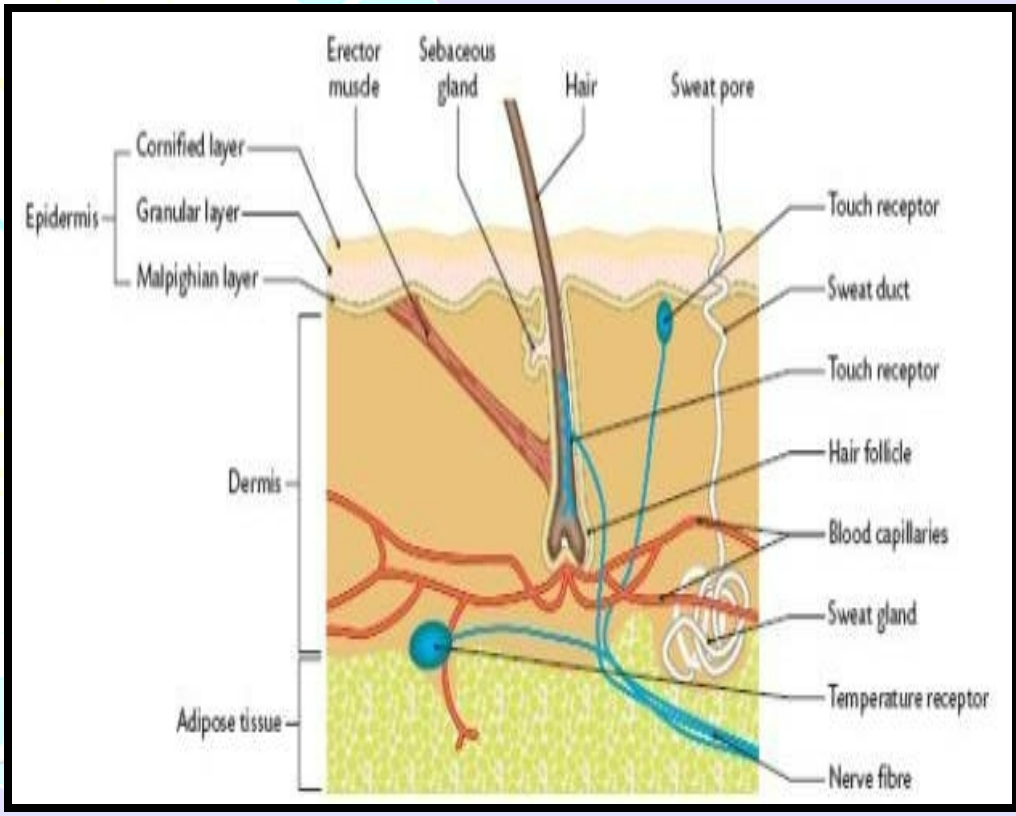
- Composed of a thin **papillary** and a thicker **reticular** layer.

- Dermis is much thicker than the epidermis and may exceed 4mm on the sloes and palms.



The Dermis

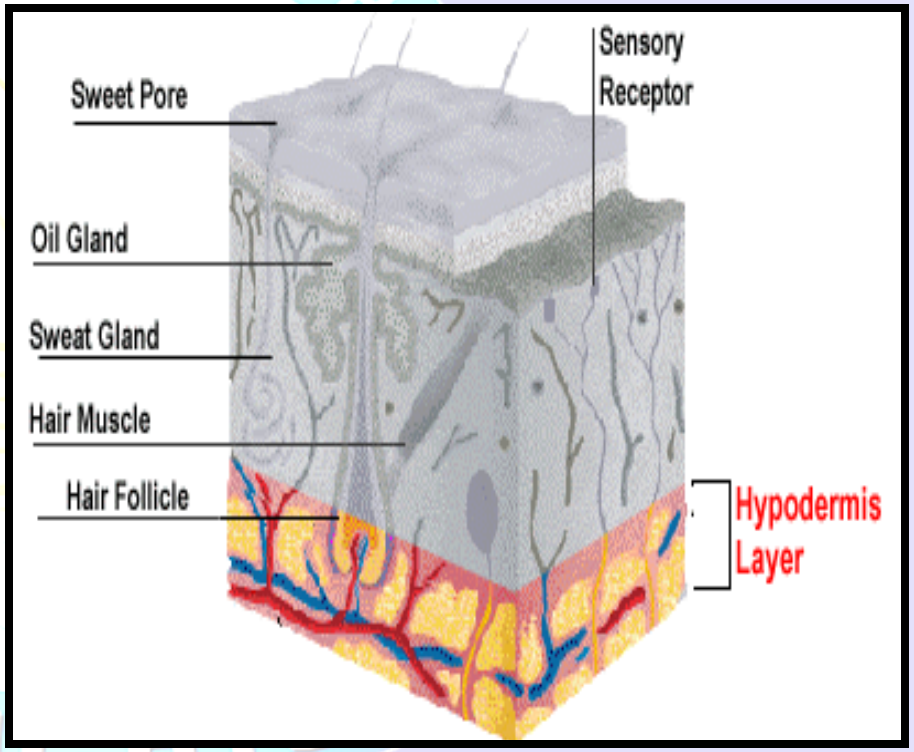
- Somatic sensory receptors process sensory information such as pain, pressure, touch and temperature.
- A variety of muscle fibers, hair follicles, sweat and sebaceous glands and many blood vessels extend at various levels of the dermis.
- Rich vascular supply of the dermis plays a critical role in the regulation of body temperature.



Hypodermis

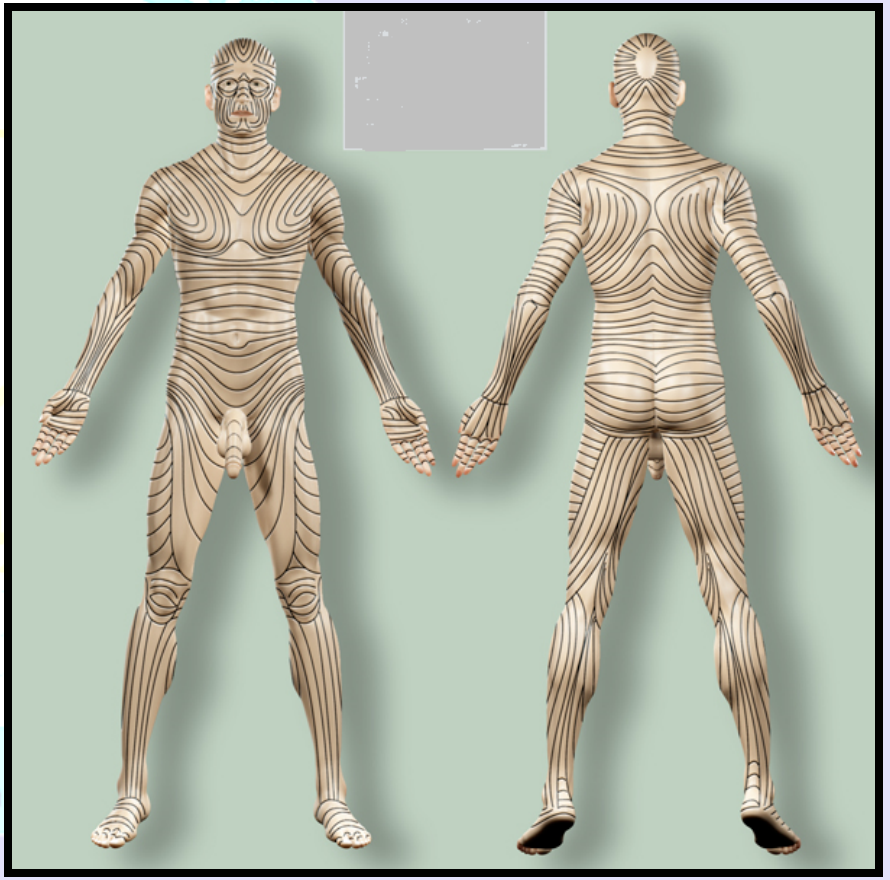
(Subcutaneous Layer, Or Superficial Fascia)

- Not part of the skin, but lies deep to the dermis thus forms a connection between skin and underlying structures.
- Hypodermis is mostly loose fibrous and adipose tissue.
- Fat content of the hypodermis varies with the state of nutrition and in obese individuals may exceed 10 cm in thickness in areas.



Tension lines of skin (cleavage lines or langer lines)

- Are due to due to the patterns of arrangements of collagen fibers in the dermis.
- Tends to spiral longitudinally over limbs and run transversely in neck and trunk.
- Incision made along the skin tension lines heal with minimum scaring.



SKIN COLOR

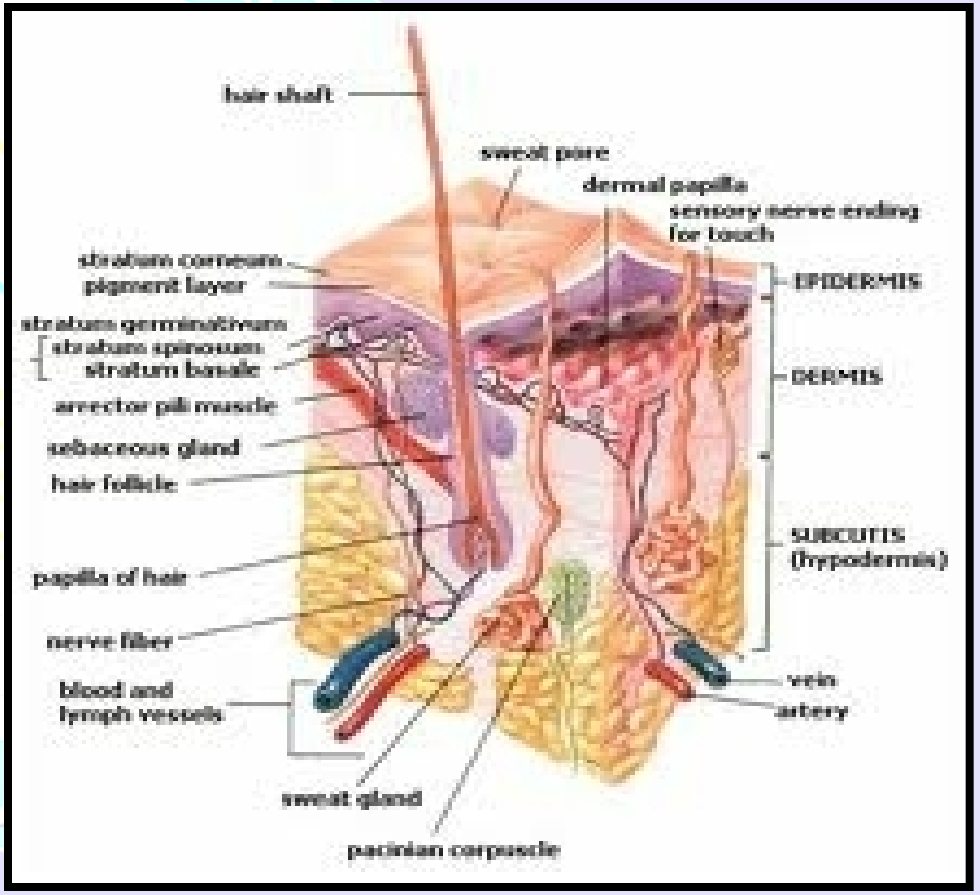
- Main determinant of skin color is the quantity of **melanin** deposited in the cells of the epidermis.
- Albino individuals (Albinism) have a characteristic absence of the pigment in their hair, skin and eyes.
- Yellow pigment beta – carotene found in many vegetables and fruits also contribute to skin color.



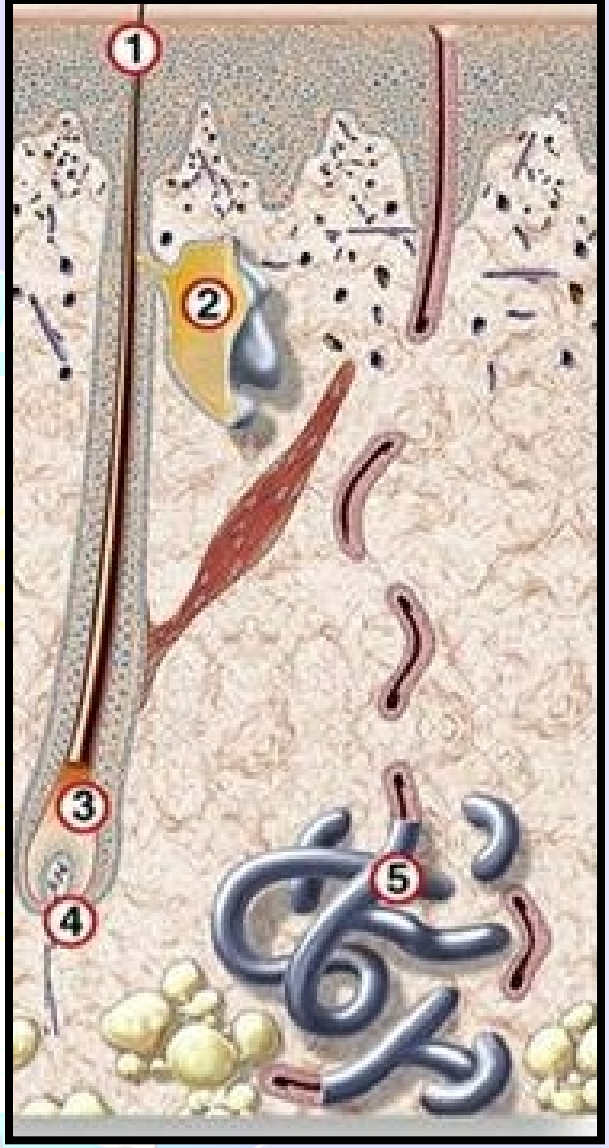
Skin appendages

- **Skin appendages** are appendages that are associated with the skin and serve a particular function.

- Sweat glands, hair, nails and sebaceous glands are all considered epidermal appendages.



- **Hairs** (sensation, heat loss, filter for breathing, protection).
- **Arrector pilli** (smooth muscles that pull hairs straight).
- **Sebaceous glands** (secrete sebum onto hair follicle to oil the hair).
- **Sweat glands** (sweat can be secreted with strong odour (apocrine) or with a faint odour (eccrine)) and **nails** (protection).



1 Hair shaft
2 Sebaceous gland
3 Bulb
4 Hair papillae
5 Sweat gland

NAILS

- Lunula is the half moon shaped white area on a nail.
- Anatomical term for the cuticle is the eponychium.
- Matrix is the region of the nails where there are dividing cells and nail growth.
- Nail plate rests on the nail bed.
- Nail root is the proximal portion of the nail that is underneath skin.



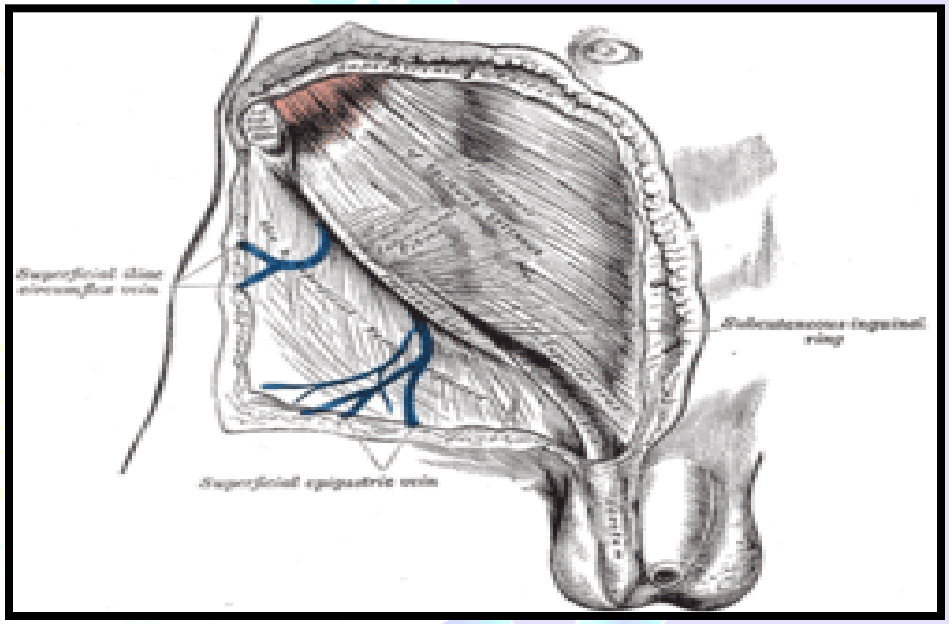
FASCIA

- **Fascias** constitute the wrapping , packing, and insulating materials of the deep structures of the body .

Superficial fascia

This is found in the subcutis in most regions of the body, blending with the reticular layer of the dermis.

E.g. fascia of scarpa.

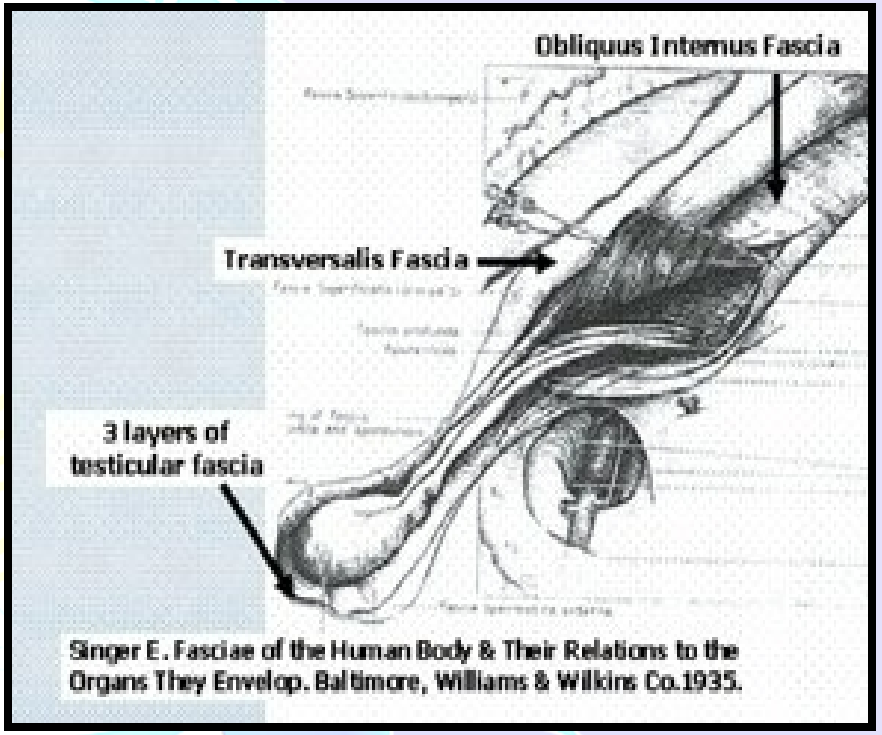


The subcutaneous inguinal ring.
(Superficial fascia visible at top.)

DEEP FASCIA

- A dense, organized connective tissue layer, devoid of fat, that covers most of the body deep to the skin and subcutaneous tissue.

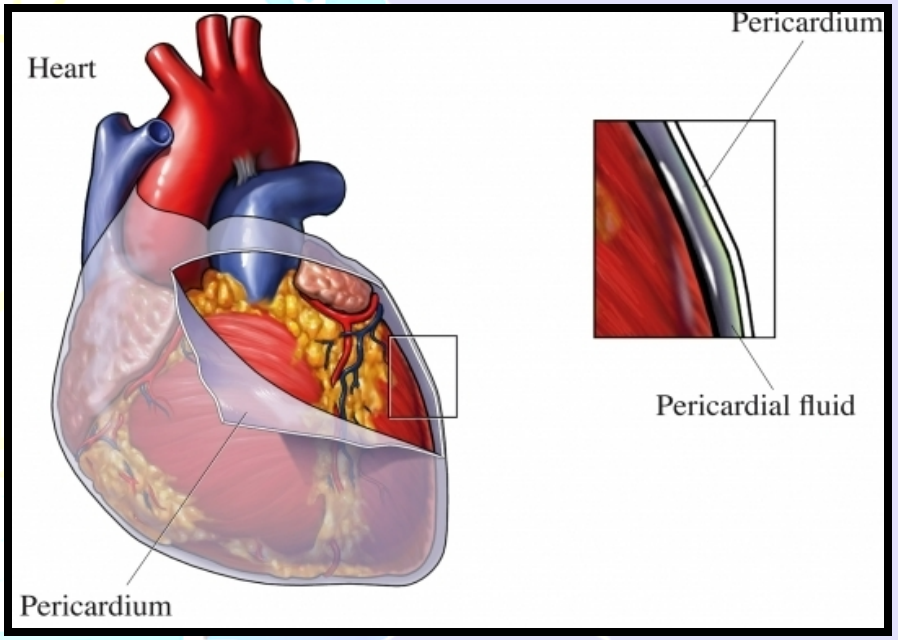
- E.g. Transversalis fascia.



Visceral Fascia

- This suspends the organs within their cavities and wraps them in layers of connective tissue membranes.

- E.g. pericardium.



Functions

- Fasciae are normally thought of as passive structures that transmit mechanical tension generated by muscular activities or external forces throughout the body.

- Function of muscle fasciae is to reduce friction to minimize the reduction of muscular force. In doing so, fasciae allow muscles to glide over each other.

Summary

- **Integumentary system:**
- Skin, consists of the epidermis, dermis, and specialized structures (hair follicles, sebaceous glands and sweat glands).

- Sweat glands, hair, nails and sebaceous glands are all considered epidermal appendages.

- Skin plays important roles in protection, heat regulation, sensation, synthesis and storage of vitamin D, features tension lines.

REFERENCES

- **Clinically oriented Anatomy.**
- **Sixth Edition**
  - Keith L. Moore.
  - Arthur F . Dally
  - Anne M. R. Agur

- **Anthony's textbook of anatomy physiology .**
- **18th edition .**



THANK YOU